

UC10 DATA FOUNDATION & FEATURE ENGINEERING TECHNICAL GUIDE

Data-First Analysis for Healthcare Pipeline Anomaly Detection

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DOCUMENT CONTROL

Field	Value
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Project	UC10 — Data Quality & Pipeline Monitoring
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Status	DATA FOUNDATION STATUS: READY TO FREEZE
Primary Data Sources	CMS Synthetic Beneficiary, FFS Claims, PDE
Primary Technologies	Python, DuckDB, scikit-learn, Parquet
Prepared For	Technical Engineering & Architecture Team
Purpose	Authoritative data and feature engineering reference for UC10

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1. Executive Data Summary

This document serves as the authoritative technical reference for the UC10 Data Foundation and Feature Engineering workflow. The UC10 pipeline is designed to detect anomalies in healthcare operational data streams using parallel detection mechanisms (Data Quality Rules, Rolling Median/MAD, and Isolation Forest).

Before building the final production simulation, the team performed a deep empirical audit of the available datasets to establish correct data relationships, resolve structural grain differences, and define an 18-feature analytical matrix. This document outlines exactly what data was investigated, why the core datasets were selected, how raw records are mathematically transformed into behavioral features, and how those features are consumed by the detection layer.

2. Data Sources Investigated

The project performed an empirical investigation of multiple publicly available healthcare data ecosystems to identify the optimal foundation for anomaly detection.

2.1 CMS Synthetic Healthcare Data (PRIMARY SOURCE)

The **CMS Synthetic Medicare Enrollment, Fee-for-Service (FFS) Claims, and Prescription Drug Event (PDE)** ecosystem was heavily investigated. This data includes synthetic Beneficiary summaries, medical activity (Claims), and pharmacy activity (PDE). It ultimately became the finalized **CORE DATA ECOSYSTEM**.

2.2 Medicare Part D Prescribers

The **CMS Medicare Part D Prescribers / Provider and Drug** dataset was investigated as an alternative. It provides provider-level and drug-level summary metrics. It was **NOT** selected as the primary active pipeline dataset due to its high level of aggregation, the absence of a **BENE_ID**, and its inability to support cross-dataset transactional anomaly tracing. It serves a different analytical purpose and is reserved for optional contextual/reference data.

2.3 Healthcare.gov PUF

The **Healthcare.gov Public Use Files (PUF)** (including **plan_attributes_PUF**, **plan_id_crosswalk_PUF**, and **service_area_PUF**) were investigated. They are strictly **supporting/optional** reference sources and do not contain transactional healthcare activity.

2.4 DQ Reference Documentation

The public documentation for **Great Expectations** and **Soda** were investigated. They were not treated as healthcare transaction data, but were used to inform Data Quality rule design, providing examples of completeness, uniqueness, validity, and consistency checks.

3. Dataset Selection Decision

WHY WE CHOSE THE CMS SYNTHETIC ECOSYSTEM

Dataset Option	Beneficiary Link	Claim/Event Grain	Cross-Dataset Analysis	DQ Analysis	Anomaly Detection	Final Role
A. Beneficiary + FFS Claims + PDE	Yes (BENE_ID)	Yes (Line/Event)	High	High	High	Core Pipeline

Dataset Option	Beneficiary Link	Claim/Event Grain	Cross-Dataset Analysis	DQ Analysis	Anomaly Detection	Final Role
B. Beneficiary + Part D Prescribers	No	No (Aggregated)	Low	Low	Low	Optional
C. PUF / Supporting Data	No	No	Low	Low	Low	Supporting

DESIGN DECISION Architecture A (Beneficiary + FFS Claims + PDE) was selected because it provides strong relational connectivity via **BENE_ID**, transaction/event-level detail, and the temporal granularity required for cross-dataset anomaly tracing and Isolation Forest modeling.

4. Core Dataset Description

The core UC10 data ecosystem consists of three empirical datasets:

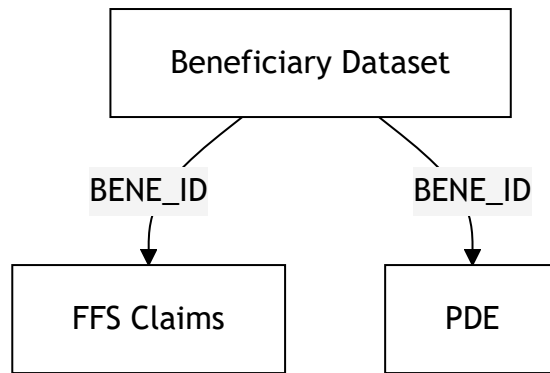
BENEFICIARY - Purpose: Member reference and master dataset. - **Grain:** 1 row = 1 unique beneficiary. - **Key:** BENE_ID - **Size:** 5,975 rows.

FFS CLAIMS - Purpose: Medical activity and institutional/non-institutional claims. - **Grain:** Claim-line level. - **Keys:** CLM_ID + CLM_LINE_NUM. Linked via BENE_ID. - **Size:** 1,121,004 rows (representing 90,705 unique claims across 7,971 beneficiaries).

PDE - Purpose: Pharmacy and prescription drug events. - **Grain:** Prescription-event level. - **Keys:** PDE_ID. Linked via BENE_ID. - **Size:** 515,520 rows (across 7,403 beneficiaries).

5. Dataset Relationships

The core architecture relies on a centralized relational hub:



Relationship Mechanics: - The BENE_ID is the central referential key linking Claims and PDE back to the Beneficiary master table. - Claims and PDE have a one-to-many relationship with the Beneficiary. - **Orphan Records:** The empirical audit found orphan records (Claims or PDEs containing a BENE_ID not present in the Beneficiary table). These are not removed; they are highly valuable because they become explicit **Data Quality (DQ) anomaly evidence**.

6. Grain and Temporal Analysis

The datasets possess inherently different structural grains: - **Beneficiary:** 1 row = 1 beneficiary - **Claims:** 1 row = 1 claim line (multiple lines per claim encounter) - **PDE:** 1 row = 1 prescription event

Because 100 claim lines does not equal 100 prescription events, these records cannot be compared directly.

Temporal Aggregation Strategy: The feature engineering process normalizes the data into a **Common Analytical Window** (Daily). By transforming raw records into daily volumes and medians, the pipeline prevents grain-driven distortion. Furthermore, a rolling historical baseline is used to compare "today's" behavior against "recent historical" behavior, inherently accounting for natural weekend/weekday volume volatility.

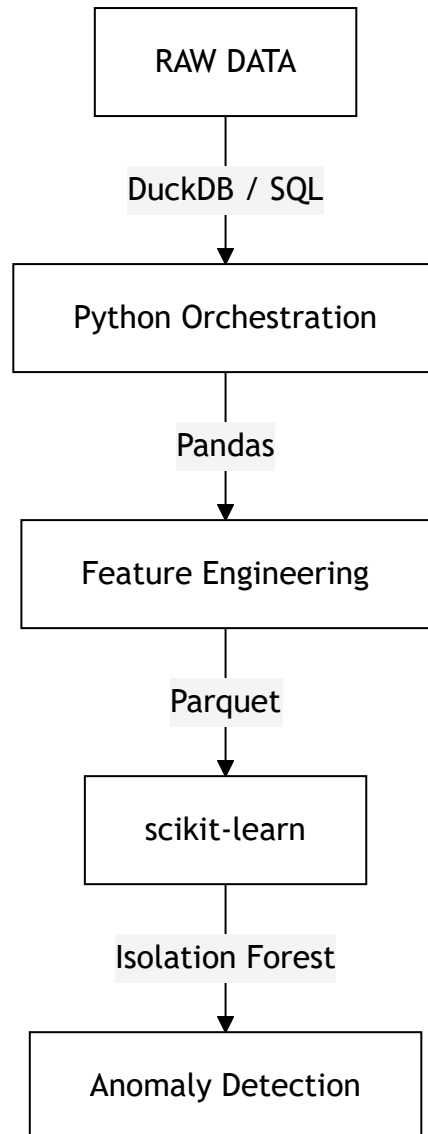
7. Data Quality Analysis

Data Quality (DQ) in UC10 is not merely preprocessing. **Data quality problems become anomaly evidence.**

The audit investigated: - **Completeness:** Missing identifiers and required fields. - **Uniqueness:** Duplicate primary keys (CLM_ID + CLM_LINE_NUM or PDE_ID). - **Validity:** Invalid or logically impossible dates. - **Referential Integrity:** Cross-dataset mismatches (e.g., orphans).

These dimensions are captured by the DQ Engine, resulting in violation rates that are directly fused into the final anomaly assessment.

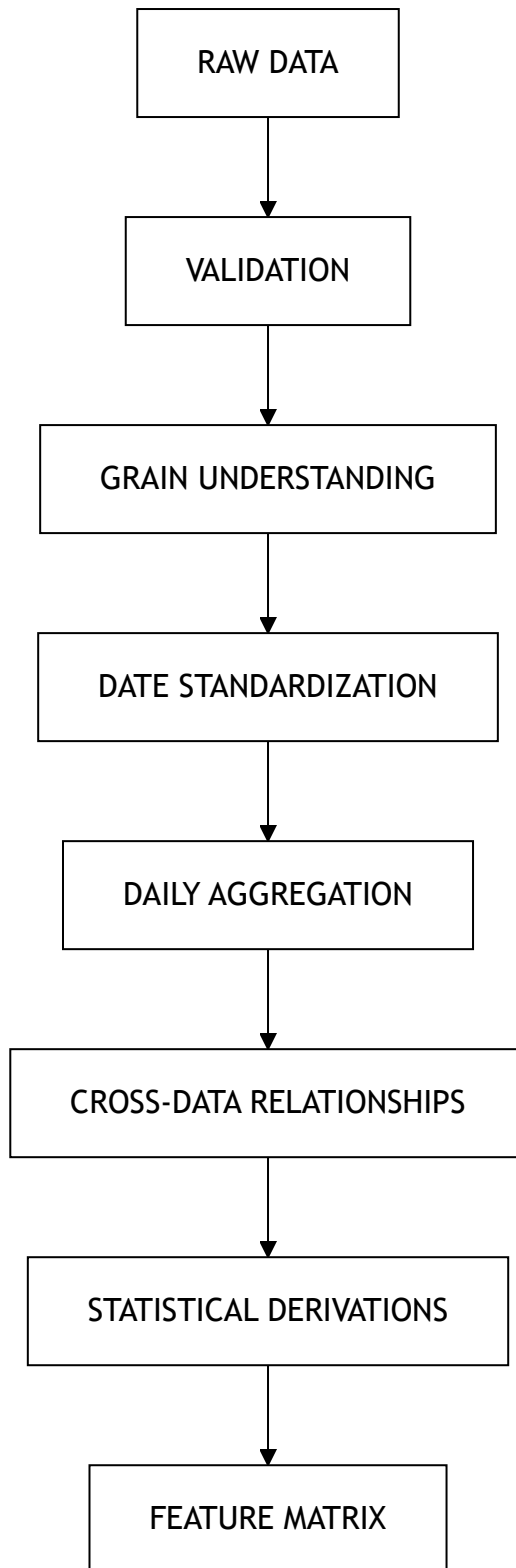
8. Technology Stack



Technologies: - **Python:** Primary orchestration, analysis, and feature engineering language. - **DuckDB & SQL:** High-performance SQL analysis over large CSV datasets without requiring the entire dataset to be loaded into memory. - **Pandas:** Targeted dataframe transformations and validations. - **Parquet:** Efficient, strongly-typed storage of the resulting feature matrix. - **scikit-learn:** Statistical and ML processing (Isolation Forest). - **YAML:** Configurable DQ rules and anomaly parameters. - **Markdown:** Auditable technical documentation.

9. Feature Engineering

The detection models do NOT consume raw healthcare records directly. They consume a daily behavioral representation.



Feature engineering collapses millions of heterogeneous rows into a standardized 18-feature matrix per day, ensuring mathematically sound inputs for the Isolation Forest and Statistical detectors.

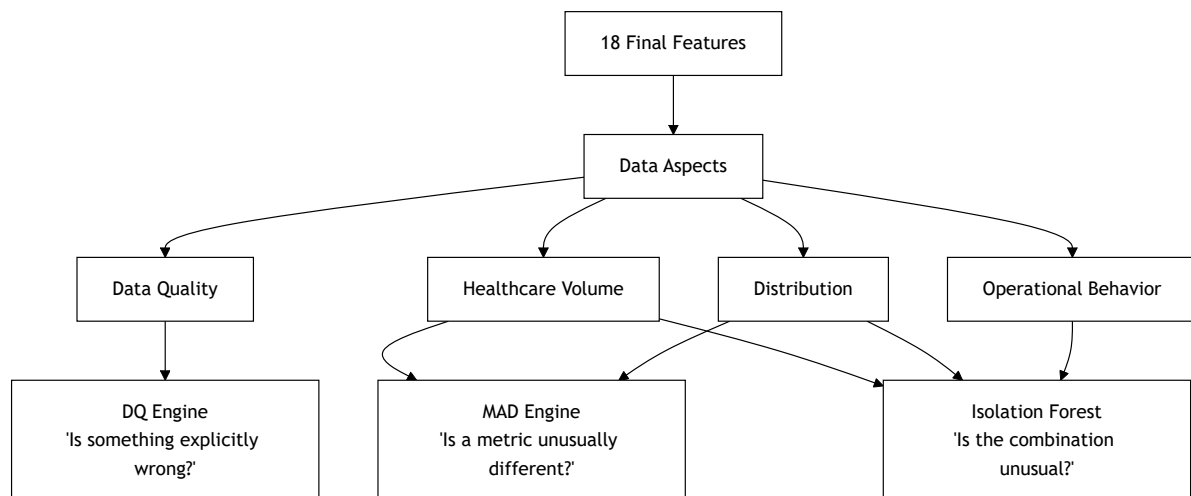
10. 18-Feature Dictionary

The complete, authoritative mapping of all 18 features used in the UC10 pipeline.

Feature Name	Source	What It Measures	Data Aspect	Detector	Why It Matters
null_rate	Derived	Rate of missing critical identifiers	Data Quality	DQ Rules	Indicates upstream data corruption
duplicate_rate	Derived	Rate of duplicate primary keys	Data Quality	DQ Rules	Indicates replay or system retry errors
claims_null_rate	Derived	Missingness specifically in Claims	Data Quality	DQ Rules	Claim-specific data corruption
cross_dataset_mismatch_rate	Derived	Orphan records missing BENE_ID master	Data Quality	DQ Rules	Referential integrity failure
dq_violation_rate	Derived	Overall DQ failure rate	Data Quality	DQ Rules	Aggregate quality health
pde_count	PDE	Daily prescription volume	Healthcare Volume	MAD, IF	Baseline pharmacy activity
claim_count	FFS Claims	Daily medical claim volume	Healthcare Volume	MAD, IF	Baseline medical activity
unique_beneficiary_count	Claims + PDE	Distinct members seen today	Healthcare Volume	IF	Breadth of population impact
unique_provider_count	Claims + PDE	Distinct providers operating today	Healthcare Volume	IF	Breadth of provider network impact
claim_pde_ratio	Derived	Ratio of medical to pharmacy activity	Healthcare Volume	MAD, IF	Detects divergent ingestion failures
volume_change_pct	Derived	Day-over-day total volume shift	Healthcare Volume	MAD, IF	Identifies catastrophic drops/spikes
median_days_supply	PDE	Distribution of prescription duration	Distribution	MAD, IF	Detects unusual clinical coding shifts

Feature Name	Source	What It Measures	Data Aspect	Detector	Why It Matters
median_rx_cost	PDE	Distribution of prescription cost	Distribution	MAD, IF	Detects financial metric shifts
median_claim_amount	FFS Claims	Distribution of claim financial value	Distribution	MAD, IF	Detects financial claim shifts
processing_duration	Simulated	Time taken to process the batch	Operational	IF	Indicates system drag or latency
throughput	Simulated	Records processed per second	Operational	IF	Indicates processing efficiency
failure_rate	Simulated	System-level processing errors	Operational	IF	Hardware or infrastructure errors
backlog	Simulated	Workload pending execution	Operational	IF	Identifies queuing and SLA risk

11. Feature → Aspect → Detector



Different anomalies require different mechanisms. **DQ** strictly catches boolean violations. **MAD** identifies univariate volatility. **Isolation Forest** evaluates the multivariate combination of features to catch complex structural shifts.

12. Controlled Anomaly Scenarios

To validate the architecture, anomalies were injected into **TESTING DATA** (not the clean baseline).

1. **Missing Identifier:** Affects `null_rate`. Caught by **DQ**. Evidence: Sharp spike in null percentage.
 2. **Duplicate Record:** Affects `duplicate_rate`. Caught by **DQ**. Evidence: PK duplication.
 3. **Invalid Date:** Affects `dq_violation_rate`. Caught by **DQ**. Evidence: Out-of-bounds dates.
 4. **Referential Integrity Failure:** Affects `cross_dataset_mismatch_rate`. Caught by **DQ**.
 5. **Volume Drop:** Affects `claim_count / pde_count`. Caught by **MAD**. Evidence: Metric drops below 3 MAD threshold.
 6. **Multivariate Degradation:** Affects multiple metrics subtly (e.g. `claim_pde_ratio` shifts alongside `processing_duration`). Caught by **Isolation Forest**.
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13. Real vs Derived vs Simulated Data

The dataset features fall into three strict categories:

CATEGORY A — DIRECTLY OBSERVED DATA Real CMS data attributes: Claims, PDE, Beneficiary identifiers, service dates, claim amounts, drug measures.

CATEGORY B — DERIVED FEATURES Mathematically calculated from real data: `claim_count`, `pde_count`, ratios, volume change, mismatch rates, medians.

CATEGORY C — SIMULATED OPERATIONAL TELEMETRY Operational fields: `processing_duration`, `throughput`, `failure_rate`, `backlog`. *Note: CMS datasets provide healthcare activity, not live pipeline execution telemetry. These operational features are simulated for prototype purposes and must be replaced with real observability data in production.*

14. Authentication / Authorization Data Gap

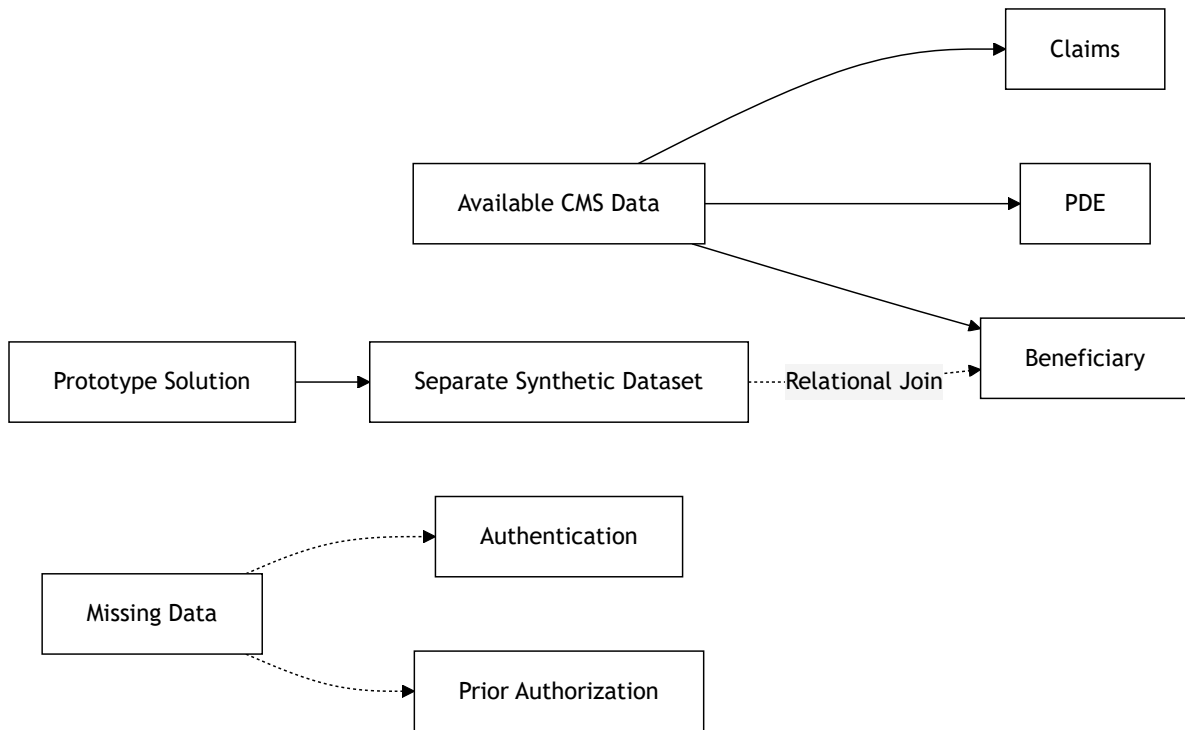
The available CMS datasets **do not** contain authentication, authorization, or prior-authorization transaction data.

Missing Elements: - Authentication event / System identity - Authorization request / Workflow status
- Approval/denial timestamp - Decision reason code

These fields cannot be directly modeled from the core CMS datasets, and must not be silently invented into the real clinical data.

15. Synthetic Authorization Approach

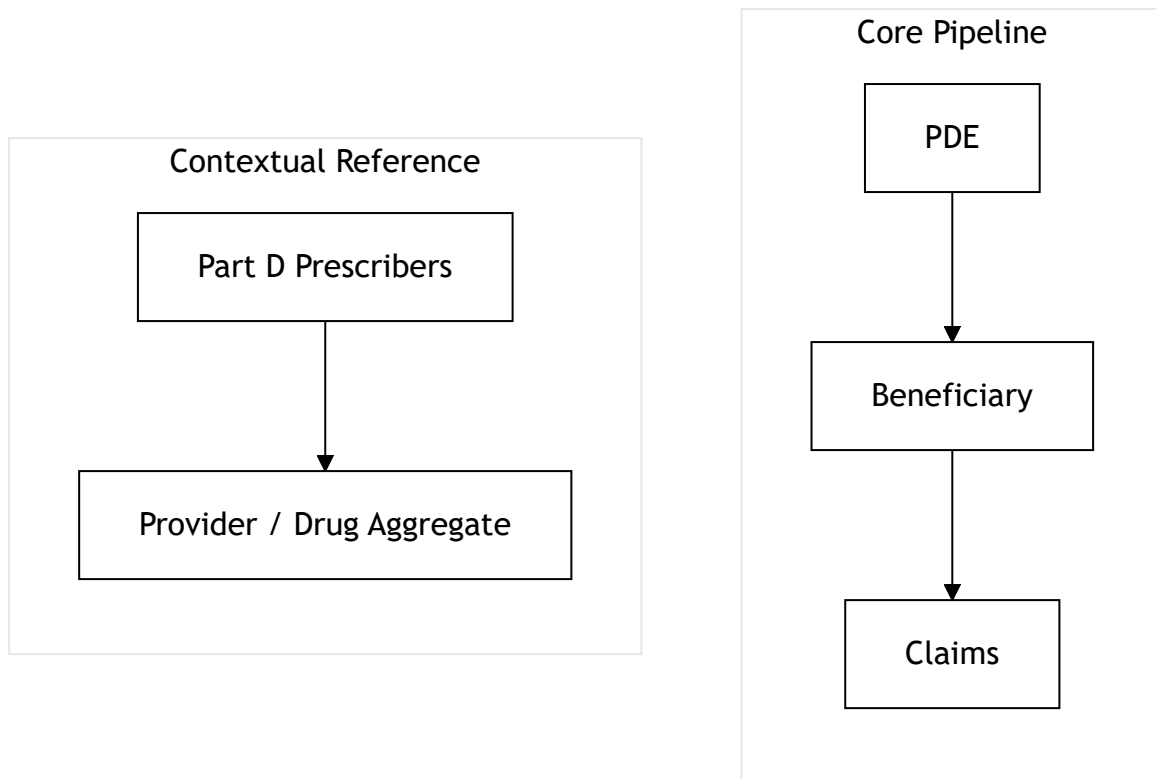
To demonstrate authorization anomalies without corrupting the core CMS data, a separate **synthetic operational dataset** is proposed.



This synthetic dataset (containing AUTH_EVENT_ID, REQUEST_TS, APPROVAL_STATUS) remains isolated and clearly labeled, allowing the prototype to demonstrate access workflow anomalies while preserving clinical data integrity.

16. Optional / Supporting Datasets

Medicare Part D Prescribers (Provider and Drug) Useful for provider-level context and aggregated pharmacy analysis. It should **not** replace PDE in the core pipeline because PDE provides event-level data and direct BENE_ID relationships essential for cross-dataset anomaly analysis.



17. Current Implementation Status

Work	Status
Data source investigation	COMPLETE
Dataset selection	COMPLETE
Relationship analysis	COMPLETE
DQ profiling	COMPLETE
Temporal analysis	COMPLETE
Feature engineering	COMPLETE
Feature validation	COMPLETE
DQ detector	COMPLETE
MAD detector	COMPLETE
Isolation Forest	COMPLETE
Controlled anomaly testing	COMPLETE
Evidence fusion	COMPLETE
Anomaly assessment	COMPLETE

Work	Status
RAG	PROTOTYPE COMPLETE
Production pipeline	NOT YET BUILT
Real operational telemetry	NOT AVAILABLE
Real authentication data	NOT AVAILABLE

Note: The prototype demonstrates detection capability but is not a fully autonomous production system.

18. Why Data-First Sequencing

The project intentionally followed a strict sequence: DATA UNDERSTANDING & rarr; FEATURE ENGINEERING & rarr; ANOMALY DETECTION & rarr; RAG & rarr; PIPELINE DEMONSTRATION

If the data foundation is mathematically or relationally flawed, the anomaly detector and RAG layer will generate false alerts. The team established correct datasets, relationships, grain, and features before advancing to production orchestration.

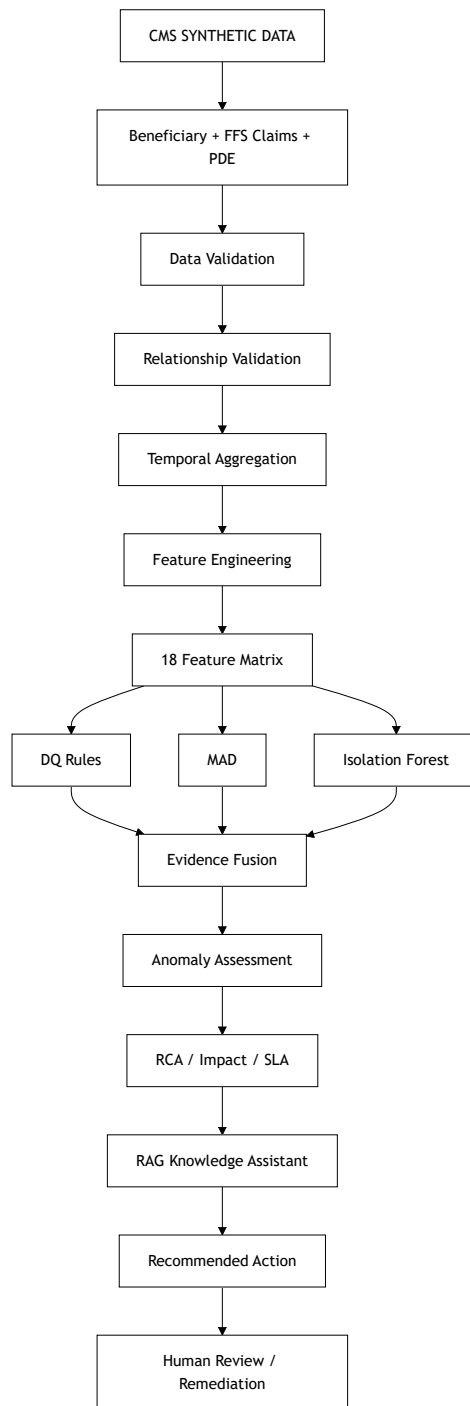
19. Remaining Development

COMPLETED DATA WORK: Dataset selection, profiling, feature engineering, and core anomaly detection logic.

NEXT DEVELOPMENT WORK: - Pipeline simulation/prototype execution - Integration of data ingestion and processing - Connecting detection engines to the simulated pipeline - Synthetic authentication/authorization demonstration - Human-in-the-loop remediation UI - Production-like orchestration

Note: RAG remains a knowledge-assistant layer and must not be described as the component that detects anomalies.

20. Final Data Architecture



21. Data Source References

[1] CMS — Synthetic Medicare Enrollment, FFS Claims and PDE
<https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable->

Public-Use-Files/SynPUFs/DE_Syn_PUF

[2] CMS — Medicare Part D Prescribers by Provider and Drug
<https://data.cms.gov/provider-summary-by-type-of-service/medicare-part-d-prescribers>

[3] Healthcare.gov — Public Use Files
<https://www.healthcare.gov/health-and-dental-plan-datasets-for-researchers-and-issuers/>

[4] Great Expectations Documentation
<https://docs.greatexpectations.io/>

[5] Soda Documentation
<https://docs.soda.io/>

22. Final Takeaway

The UC10 Data Foundation safely converts heterogeneous clinical records into a standardized daily behavioral view. The resulting features provide distinct types of evidence (data quality, distribution, volume, and simulated operations). Each evidence type is routed to the mathematical detector best suited to identify that specific anomaly class.

DATA FOUNDATION STATUS: READY TO FREEZE